

Here are some conventions for relations on X (i.e. subsets of $X \times X$).

- If $R \subseteq X \times X$ then we shorthand $(a, b) \in R$ as aRb .
- We write $\Delta X := \{(a, a) : a \in X\}$. Note that $\Delta X \subseteq R$ if and only if R is a reflexive relation.
- Given $R \subseteq X \times X$ we define $R^{-1} := \{(a, b) : (b, a) \in R\}$. Note that $R^{-1} \subseteq R$ if and only if R is a symmetric relation.
- Given $R, S \subseteq X \times X$, we define $R \circ S \subseteq X \times X$ where $aR \circ Sb$ iff there exists $c \in X$ such that aRc and cSb . Note that $R \circ R \subseteq R$ if and only if R is a transitive relation.

Remark 1: If $R_1, R_2, R_3 \subseteq X \times X$, then $(R_1 \circ R_2) \circ R_3 = R_1 \circ (R_2 \circ R_3)$.

Proof:

If $a(R_1 \circ R_2) \circ R_3b$ then there exists c with $a(R_1 \circ R_2)c$ and cR_3b . Next, there exists d with aR_1d and dR_2c . But now $dR_2 \circ R_3b$. So, $aR_1 \circ (R_2 \circ R_3)b$. This proves that $(R_1 \circ R_2) \circ R_3 \subseteq R_1 \circ (R_2 \circ R_3)$. Mirror reasoning shows the other inclusion. ■

Remark 2: If $R, S \subseteq X \times X$ then $(R \circ S)^{-1} = S^{-1} \circ R^{-1}$.

Proof: $a(R \circ S)^{-1}b$ iff $bR \circ Sa$. But that happens iff there exists c such that bRc and cSa , and that happens iff there exists c such that $aS^{-1}c$ and $cR^{-1}b$. That is equivalent to saying that $aS^{-1} \circ R^{-1}b$. ■

Remark 3: If $S, R \subseteq X \times X$ with $\Delta X \subseteq S$ then $R \subseteq R \circ S$ and $R \subseteq S \circ R$.

Why?

If aRb then aSa implies that $aS \circ Rb$. Similarly bSb implies that $aR \circ Sb$.

Remark 4: If $R, R', S \subseteq X \times X$ and $R \subseteq R'$ then $S \circ R \subseteq S \circ R'$ and $R \circ S \subseteq R' \circ S$.

Hopefully this is obvious.

For our purposes here, the relations on X that we will care about will be relations giving a way of describing when any two elements a and b in X are "close" to each other.

A uniform structure on a set X is a filter $\mathcal{U}$ on $X \times X$ satisfying that:

- $\Delta X \subseteq D$ for all $D \in \mathcal{U}$
- $D \in \mathcal{U} \implies D^{-1} \in \mathcal{U}$
- If $D \in \mathcal{U}$ then there exists $D' \in \mathcal{U}$ with $D' \circ D' \subseteq D$.

A uniform space $(X, \mathcal{U})$ is a set X equipped with a uniform structure $\mathcal{U}$.

Two trivial uniform structures are as follows:

1. The discrete uniform structure on a set X is the filter consisting of all sets containing ΔX .

2. The indiscrete uniform structure on a set X is the filter whose only element is $X \times X$.

In order to get more interesting uniform structures, we consider uniform structures generated by collections of sets. Unfortunately, the third bullet point in the definition of a uniform structure makes it so that an intersection of many uniform structures may not be a uniform structure. Hence, we can't always define a "smallest" uniform structure generated by some collection of sets in the same way that we can for topologies. Something we can do though is consider a filterbase for a uniform structure. Then a smallest uniform structure containing that filterbase is guaranteed to exist as we will see.

We say a uniformity base on a set X is a filterbase $\mathcal{B}$ on $X \times X$ satisfying that:

- $\Delta X \subseteq E$ for all $E \in \mathcal{B}$
- $E \in \mathcal{B} \implies E^{-1} \in \mathcal{B}$
- If $E \in \mathcal{B}$ then there exists $E' \in \mathcal{B}$ with $E' \circ E' \subseteq E$.

Claim: The filter $\mathcal{U}$ generated by a uniformity base $\mathcal{B}$ is a uniform structure.

Proof:

Let $D \in \mathcal{U}$. Then there exists $E \in \mathcal{B}$ with $E \subseteq D$. Since $\Delta X \subseteq E$, we can conclude $\Delta X \subseteq D$ as well. Also, note that $E \subseteq D \implies E^{-1} \subseteq D^{-1}$. And since $E^{-1} \in \mathcal{B}$ we know $D^{-1} \in \mathcal{U}$. Finally, let $E' \in \mathcal{B}$ satisfy that $E' \circ E' \subseteq E$. Then we also have that $E' \in \mathcal{U}$ and $E' \circ E' \subseteq D$. ■

Here are two* important uniform structures defined using uniformity bases.

1. Suppose (X, ρ) is a metric space. Then we define $\mathcal{B} := \{D_\varepsilon : \varepsilon > 0\}$ where $D_\varepsilon := \{(x, y) \in X \times X : \rho(x, y) < \varepsilon\}$. Next, we let the metric uniform structure $\mathcal{U}$ be generated $\mathcal{B}$.

$\mathcal{B}$ is trivially directed by inclusion since we can make a stronger statment that $\{D_\varepsilon\}_{\varepsilon>0}$ is ordered by inclusion. Thus, $\mathcal{B}$ is a filterbase. Furthermore, $\rho(x, x) = 0$ for all $x \in X$ guarentees that $\Delta X \subseteq D_\varepsilon$ for all ε . Similarly, $\rho(x, y) = \rho(y, x)$ guarentees that $D_\varepsilon = D_\varepsilon^{-1}$ for all ε .

Finally, let $C \geq 1$ satisfy that $\rho(x, z) \leq C(\rho(x, y) + \rho(y, z))$ for all $x, y, z \in X$. Then we can see that $D_{\frac{\varepsilon}{2C}} \circ D_{\frac{\varepsilon}{2C}} \subseteq D_\varepsilon$.

I used an arbitrary constant C instead of just setting $C = 1$ cause I wanted to show that this construction also works for certain semimetrics (such as quasi-norms).

2. Suppose $(G, \mathcal{T})$ is a topological group. We define the left uniform structure $\mathcal{U}_\ell$ to be generated by the collection $\mathcal{B}$ of sets of the form $L_N := \{(x, y) \in G \times G : y^{-1}x \in N\}$ where N is a neighborhood of the identity element e of G .

In other words, xL_Ny iff $x \in yN$.

To see why $\mathcal{B}$ is a filterbase, first note that $G \times G \in \mathcal{B}$ always. Hence, $\mathcal{B}$ is nonempty. Also, since N is a neighborhood of the identity, we know that $x = xe \in xN$ for

all $x \in G$. Hence, $\Delta X \subseteq L_N$ for all neighborhoods N of the identity. This also means $\emptyset \notin \mathcal{B}$. Finally, consider L_{N_1} and L_{N_2} in $\mathcal{B}$. Then $N := N_1 \cap N_2$ is still a neighborhood of e . Also, $x \in yN \implies x \in yN_1$ and $x \in yN_2$. So, $L_N \subseteq L_{N_1} \cap L_{N_2}$. And this proves that $\mathcal{B}$ is directed by inclusion.

As for additionally showing that $\mathcal{B}$ is a uniformity base, note that if N is an open neighborhood of e then so is N^{-1} . Also $x \in yN$ iff $y^{-1}x \in N$ iff $x^{-1}y \in N^{-1}$ iff $y \in xN^{-1}$. So $L_N^{-1} = L_{N^{-1}} \in \mathcal{B}$.

Next suppose N is a neighborhood of e . Then recall from [page 333](#) that we can find a neighborhood N' of e with $N'N' \subseteq N$. But now if $x \in yN'$ and $y \in zN'$, we have that $x \in zN'N' \subseteq zN$. So $L_{N'} \circ L_{N'} \subseteq L_N$.

Similarly, we define the right uniform structure $\mathcal{U}_r$ to be generated by the collection $\mathcal{B}$ of sets of the form $R_N := \{(x, y) : xy^{-1} \in N\}$ where N is a neighborhood of the identity element e .

In other words, xR_Ny iff $x \in Ny$.

Let $(X, \mathcal{U})$ be a uniform space. Then we define a relation $R := \bigcap_{D \in \mathcal{U}} D$.

- Note that R is reflexive since $\Delta X \subseteq D$ for all $D \in \mathcal{U}$.
- Also note that if xRy then xDy for all $D \in \mathcal{U}$. Yet we also know $D \in \mathcal{U}$ implies that $D^{-1} \in \mathcal{U}$. So, $xRy \in R \implies xD^{-1}y$ for all $D \in \mathcal{U}$. So, yDx for all $D \in \mathcal{U}$. In other words, xRy implies yRx . So R is symmetric.
- Finally, suppose xRy and yRz . Then for any fixed $D \in \mathcal{U}$ we can pick $D' \in \mathcal{U}$ with $D' \circ D' \subseteq D$. Yet now note that $xD'y$ and $yD'z$. So, xDz . And since D was arbitrary, we've shown that xRz . So, R is transitive.

As a result of the prior reasoning, we know R is an equivalence relation on X . We say $(X, \mathcal{U})$ is a separated uniform space if $\Delta X = R$.

Intuitively, $(X, \mathcal{U})$ is separated iff for all distinct $x, y \in X$ we can find some measure of "closeness" in $\mathcal{U}$ for which x is not close to y . As we will see once we connect uniform structures to topology, a uniform space being separated is equivalent to it being Hausdorff as a topological space.

Let $\phi : X \rightarrow Y$ be a map between two uniform spaces $(X, \mathcal{U}_X)$ and $(Y, \mathcal{U}_Y)$. We denote $\phi \times \phi : X \times X \rightarrow Y \times Y$ to be the map $(x_1, x_2) \mapsto (\phi(x_1), \phi(x_2))$. Then, we say ϕ is uniformly continuous if $(\phi \times \phi)^{-1}(E) \in \mathcal{U}_X$ for all $E \in \mathcal{U}_Y$.

Remark 1: Let $\phi : X \rightarrow Y$ and $\psi : Y \rightarrow Z$ be uniformly continuous maps between uniform spaces. Then $\psi \circ \phi$ is a uniformly continuous map from X to Z .

Remark 2: Any constant function on a uniform space X is uniformly continuous. Similarly, $\text{Id}_X : X \rightarrow X$ is always uniformly continuous.

Both remarks are trivial to verify.

Proposition: If $\phi : (X, \mathcal{U}_X) \rightarrow (Y, \mathcal{U}_Y)$ is a map between uniform spaces and $\mathcal{B}_Y$ is a uniformity base for $\mathcal{U}_Y$, then $\phi : X \rightarrow Y$ is uniformly continuous iff $(\phi \times \phi)^{-1}(E) \in \mathcal{U}_X$ for all $E \in \mathcal{B}_Y$.

($\implies$)

This direction is trivial from the fact that $\mathcal{B}_Y \subseteq \mathcal{U}_Y$.

($\impliedby$)

Consider any $D \in \mathcal{U}_Y$ and then pick $E \in \mathcal{B}_Y$ with $E \subseteq D$. Then

$(\phi \times \phi)^{-1}(D) \supseteq (\phi \times \phi)^{-1}(E) \in \mathcal{U}_X$. So, $(\phi \times \phi)^{-1}(D) \in \mathcal{U}_X$ since $\mathcal{U}_X$ is a filter. ■

Corollary: If (X, ρ_X) and (Y, ρ_Y) are metric spaces equipped with their metric uniform structures, then our new definition of uniform continuity agrees with our old definition.

Why?

Let $\mathcal{U}_X$ and $\mathcal{U}_Y$ denote the uniform structures on X and Y respectively, and suppose $\phi : X \rightarrow Y$ is a map. Now for any E_ε in the standard uniformity base for $\mathcal{U}_Y$ we have that $(\phi \times \phi)^{-1}(E_\varepsilon) \in \mathcal{U}_X$ iff there exists D_δ in the standard uniformity base for $\mathcal{U}_X$ such that $D_\delta \subseteq (\phi \times \phi)^{-1}(E_\varepsilon)$. Equivalently this means that ϕ is uniformly continuous according to our new definition iff for all $\varepsilon > 0$ there exists $\delta > 0$ with $\rho_X(x, y) < \delta \implies \rho_Y(\phi(x), \phi(y)) < \varepsilon$.

One related headache to mention is as follows:

Corollary 2: Let G be a topological group equipped with the left uniform structure $\mathcal{U}_L$, and also let $\mathbb{R}$ be equipped with the (Euclidean) metric uniform structure. Then left uniform continuity by our new definition is equivalent to Folland's notion of right uniform continuity (see [page 334](#)).

($\implies$)

Let $\phi : G \rightarrow \mathbb{R}$ be a map. Then for any $\varepsilon > 0$ we know that $E_\varepsilon := \{(x, y) : |x - y| < \varepsilon\}$ satisfies that $(\phi \times \phi)^{-1}(E_\varepsilon) \in \mathcal{U}_L$. Hence, there exists a filter $L_N \in \mathcal{U}_L$ satisfying that $L_N \subseteq (\phi \times \phi)^{-1}(E_\varepsilon)$. In other words, if $z \in xN$ then $|f(z) - f(x)| < \varepsilon$. Yet this is equivalent to saying that for all $x \in G$ and $y \in N$, we have that $|f(xy) - f(x)| < \varepsilon$.

($\impliedby$)

Just reverse all the steps in the prior direction. ■

Similarly, right uniform continuity by our new definition is equivalent to Folland's notion of left uniform continuity.

The proof is almost entirely identical to the one before except for an additional step where you shrink your neighborhood N to a symmetric neighborhood.

Unfortunately, both James and Folland's notions of left vs right make sense in the context of what they were thinking about. Thus for the time being, I'm going to stick to James's convention.

A uniform equivalence $\phi : X \rightarrow Y$ (where both X and Y are uniform spaces) is a bijective function such that both ϕ and ϕ^{-1} are uniformly continuous.

Example: Suppose $\phi : G \rightarrow H$ is a continuous group homomorphism between topological groups. Then ϕ is uniformly continuous if G and H are both equipped with the left uniform structure, and also if G and H are both equipped with the right uniform structures.

I'll only show this for the left uniform structure. Suppose N is any neighborhood of the identity e_H of H . Then let $N' = \phi^{-1}N$. Because ϕ is continuous, we also know that N' is a neighborhood of the identity e_G of G . But now if $x \in yN'$, we know that $\phi(x) \in \phi(y)N$ since ϕ is a group homomorphism. Hence, $L_{N'}$ in the uniform structure on G satisfies that $L_{N'} \subseteq (\phi \times \phi)^{-1}(L_N)$. And since the L_N generate the left uniform structure on H , we've proven that ϕ is uniformly continuous. ■

As a consequence, we can also say that if $\phi : G \rightarrow H$ is a topological group isomorphism (i.e. a homeomorphism and group isomorphism), then ϕ is a left and right uniform equivalence.

One reason to care about uniform equivalences is that we can study properties that are invariant under uniform equivalences.

Proposition: If $(X, \mathcal{U}_X)$ and $(Y, \mathcal{U}_Y)$ are uniform spaces, X is separated, and $\phi : X \rightarrow Y$ is a bijective map such that ϕ^{-1} is uniformly continuous, then Y is separated as well.

Proof:

Let $R_Y := \bigcap_{E \in \mathcal{U}_Y} E$. We want to prove $R_Y = \Delta Y$. So consider any $(y_1, y_2) \in Y \times Y$ such that $y_1 \neq y_2$. Since ϕ is bijective, we know that there exists unique distinct elements $x_1, x_2 \in X$ such that $\phi(x_1) = y_1$ and $\phi(x_2) = y_2$. And now since X is separated, we can find some $D \in \mathcal{U}_X$ such that $(x_1, x_2) \notin D$.

But now set $E = (\phi \times \phi)(D)$. Since ϕ^{-1} is uniformly continuous, we know that $(\phi \times \phi)(D) \in \mathcal{U}_Y$. Furthermore, $(\phi \times \phi)^{-1}(\{(y_1, y_2)\}) = \{(x_1, x_2)\} \not\subseteq D$. So $(y_1, y_2) \notin E$. And with that, we have shown that $(y_1, y_2) \notin \Delta Y \implies (y_1, y_2) \notin R_Y$. ■

Corollary: If $(X, \mathcal{U}_X)$ and $(Y, \mathcal{U}_Y)$ are uniform spaces and $\phi : X \rightarrow Y$ is a uniform equivalence, then X is separated iff Y is separated.

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I have more to go over from *Topological and Uniform Spaces* today. For the time being I'll still be working with uniform structures as a thing defined separately from topological spaces.

Firstly, here's another convention of James for relations on a set X . Suppose $R \subseteq X \times X$. If aRb we say that a is R -related to b . Then, we define $R[a] := \{b \in X : aRb\}$. In other words, $R[a]$ denotes the set of elements that a is R -related to.

Similarly, if $S \subseteq X$ then we define $R[S] := \bigcup_{a \in S} R[a]$.

Suppose $(X, \mathcal{U})$ is a uniform space. We say that X is totally bounded if for all $D \in \mathcal{U}$ there exists a finite set $S \subseteq X$ such that $X = D[S]$.

Remark: If $\mathcal{B}$ is a filterbase for $\mathcal{U}$ then X is totally bounded iff for all $D \in \mathcal{B}$ we can find a finite set $S \subseteq X$ such that $X = D[S]$. From this observation, hopefully it is clear how this definition of total boundedness agrees with the metric space definition when we equip our metric space with the metric uniform structure.

Proposition: Suppose $(X, \mathcal{U}_X)$ and $(Y, \mathcal{U}_Y)$ are uniform spaces and $\phi : X \rightarrow Y$ is a uniformly continuous surjection. If X is totally bounded then so is Y .

Proof:

Consider any $E \in \mathcal{U}_Y$ and then let $D := (\phi \times \phi)^{-1}(E) \in \mathcal{U}_X$. Since X is totally bounded, we can find a finite set $S \subseteq X$ with $D[S] = X$. In turn, we have that $Y = E[\phi(S)]$. After all, suppose $y \in Y$. Since ϕ is surjective, there exists $x \in X$ such that $\phi(x) = y$. Yet now pick $s \in S$ such that sDx . Then $\phi(s)Ey$. ■

Corollary: If X and Y are uniform spaces, and $\phi : X \rightarrow Y$ is a uniform equivalence, then X is totally bounded iff Y is totally bounded.

One of the main reasons for caring about uniform spaces is that they let us define what it means for a sequence to be Cauchy.

Suppose $(X, \mathcal{U})$ is a uniform space and $\{x_n\}_{n \in \mathbb{N}}$ is a sequence in X . Then we say $\{x_n\}_{n \in \mathbb{N}}$ is a Cauchy sequence if:

$$\forall D \in \mathcal{U}, \exists N \in \mathbb{N} \text{ s.t. } \forall n, m \geq N, x_n D x_m$$

Remark: Like with total boundedness, if $\mathcal{B}$ is a filterbase for $\mathcal{U}$ then it suffices to check the condition for a sequence being Cauchy only for the elements in $\mathcal{B}$. With this in mind, hopefully it is clear how this definition of a sequence being Cauchy agrees with the metric space definition when we equip out metric space with the metric uniform structure.

Similarly, we say a net $\langle x_\alpha \rangle_{\alpha \in A}$ is Cauchy if:

$$\forall D \in \mathcal{U}, \exists \alpha \in A \text{ s.t. } \forall \beta, \gamma \geq \alpha, \beta D \gamma$$

Once we define a topology associated with a uniform structure, we'll be able to discuss whether a uniform space is complete or not (i.e. if all Cauchy sequences and nets converge or not). Furthermore, we'll be able to discuss taking completions of uniform spaces.

Proposition: Suppose $(X, \mathcal{U}_X)$ and $(Y, \mathcal{U}_Y)$ are uniform spaces, $\langle x_\alpha \rangle_{\alpha \in A}$ is a Cauchy net in X , and $\phi : X \rightarrow Y$ is a uniformly continuous map. Then $\langle \phi(x_\alpha) \rangle_{\alpha \in A}$ is a Cauchy net in Y .

Proof:

Let E be a relation in $\mathcal{U}_Y$.

By the way, something I've not mentioned before is that a relation in $\mathcal{U}_Y$ would normally be called an entourage. The reason I wasn't using that terminology before

is that the word "entourage" makes my mind immediately think of a neighborhood in Y as opposed to a subset of $Y \times Y$. That said, it would be helpful to not have to keep explicitly naming the uniform structure of X and Y . So, I'm going to start calling relations in uniform structures "entourages".

Since ϕ is uniformly continuous, we know that $D := (\phi \times \phi)^{-1}(E)$ is an entourage on X . In turn, as $\langle x_\alpha \rangle_{\alpha \in A}$ is a Cauchy net, we know there exists $\alpha_0 \in A$ with $x_\beta D x_\gamma$ for all $\beta, \gamma \geq \alpha_0$. And now $\phi(x_\beta) E \phi(x_\gamma)$ for all $\beta, \gamma \geq \alpha_0$. Since E was arbitrary, we can conclude that $\langle \phi(x_\alpha) \rangle_{\alpha \in A}$ is a Cauchy net in Y . ■

Proposition: Suppose X is a uniform space. Then X is totally bounded if and only if every net $\langle x_\alpha \rangle_{\alpha \in A}$ in X admits a Cauchy subnet.